Physical Chemistry (I)

Lecture 16

In the Last Class

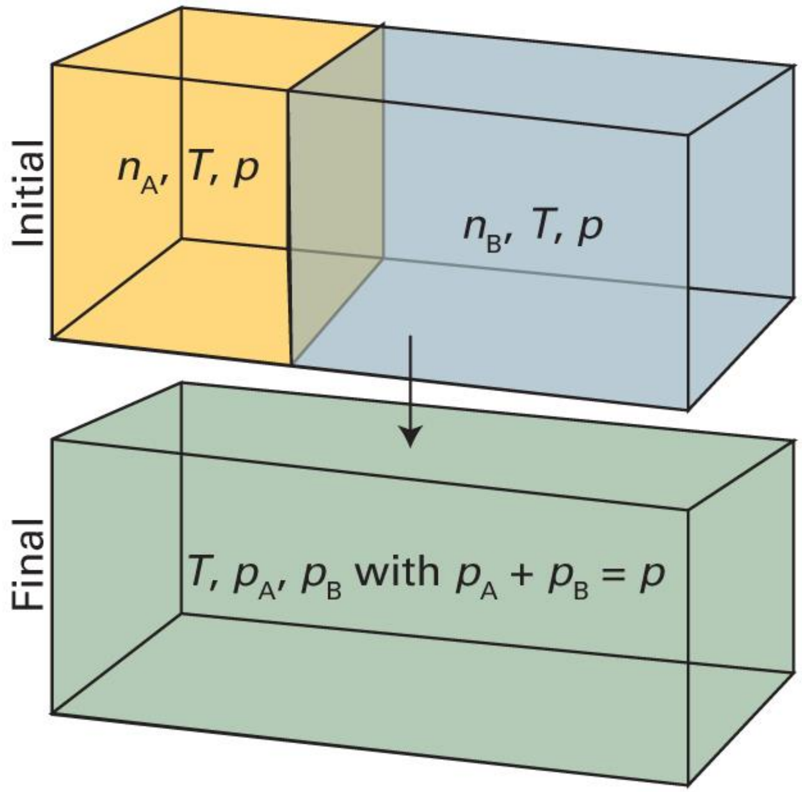
- Gas mixtures
  - Mixing of perfect gases
  - Real gas: fugacity
  - Mixing of real gases

- Liquid mixtures
  - Thermodynamics of a multi-component system
  - Partial molar quantities

Review

Mixing of Gases

(at constant T and p)

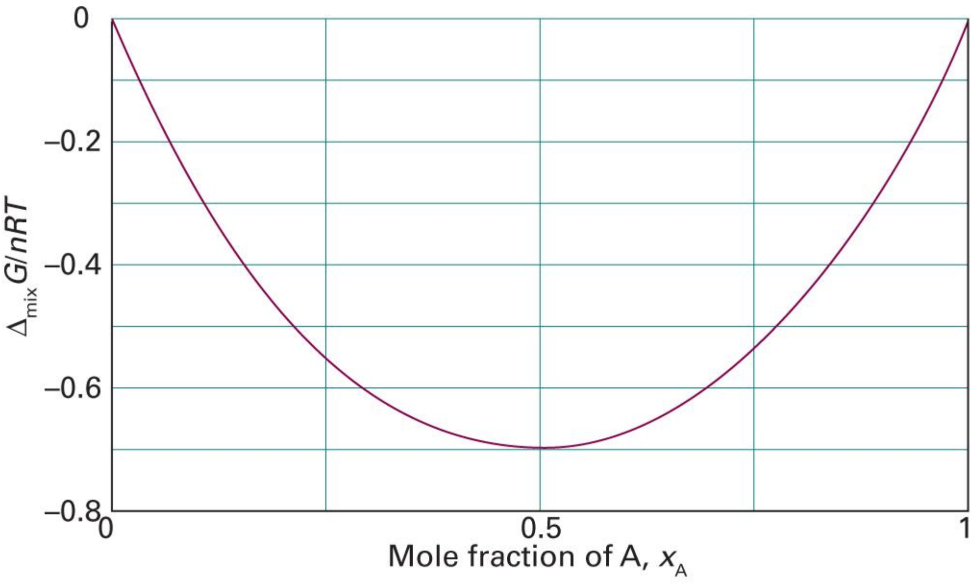


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.6)

Review

Mixing of Perfect Gases

$$\Delta_{\mathrm{mix}}G = nRT\{x_{\mathrm{A}} \ln x_{\mathrm{A}} + (1 - x_{\mathrm{A}}) \ln(1 - x_{\mathrm{A}})\}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.7)

Real Gases

For a perfect gas,

$$\mu = \mu^{\ominus} + RT \ln \frac{p}{p^{\ominus}}$$

For a real gas,

$$\mu = \mu^{\ominus} + RT \ln \frac{f}{p^{\ominus}}$$

where f is the fugacity.

Properties of Fugacity

$$f = pe^{B'p + C'p^2/2 + \cdots}$$

- As $p \to 0$, $f \to p$
- For a slightly imperfect gas,

$$f \approx pe^{B'p}$$

  - $B' > 0$ (repulsion dominant): $f > p$
  - $B' = 0$ (perfect-gas-like): $f = p$
  - $B' < 0$ (attraction dominant): $f < p$

Fugacity Coefficient

$$\phi = \frac{f}{p}$$

- For a perfect gas, $\phi = 1$
- As $p \to 0$, $\phi \to 1$
- For a slightly imperfect gas, $\phi \approx e^{B'p}$
  - $B' > 0$ (repulsion-dominant): $\phi > 1$
  - $B' < 0$ (attraction-dominant): $\phi < 1$

Mixing of Real Gases

$$\Delta_{\text{mix}}G = G_f - G_i = n_{\text{A}}RT\ln\left(x_{\text{A}}\frac{\phi_{\text{A}}}{\phi_{\text{A}}^*}\right) + n_{\text{B}}RT\ln\left(x_{\text{B}}\frac{\phi_{\text{B}}}{\phi_{\text{B}}^*}\right)$$

$$\Delta_{\text{mix}}G = nRT\left\{\underbrace{x_{\text{A}}\ln x_{\text{A}} + x_{\text{B}}\ln x_{\text{B}}}_{\text{perfect gas}} + \underbrace{x_{\text{A}}\ln\left(\frac{\phi_{\text{A}}}{\phi_{\text{A}}^*}\right) + x_{\text{B}}\ln\left(\frac{\phi_{\text{B}}}{\phi_{\text{B}}^*}\right)}_{\text{correction}}\right\}$$

In typical conditions, the correction term is much smaller and

$$\Delta_{\text{mix}}G = \Delta_{\text{mix}}H - T\Delta_{\text{mix}}S < 0$$

Review

Mixing of Liquids

When A is a pure substance,

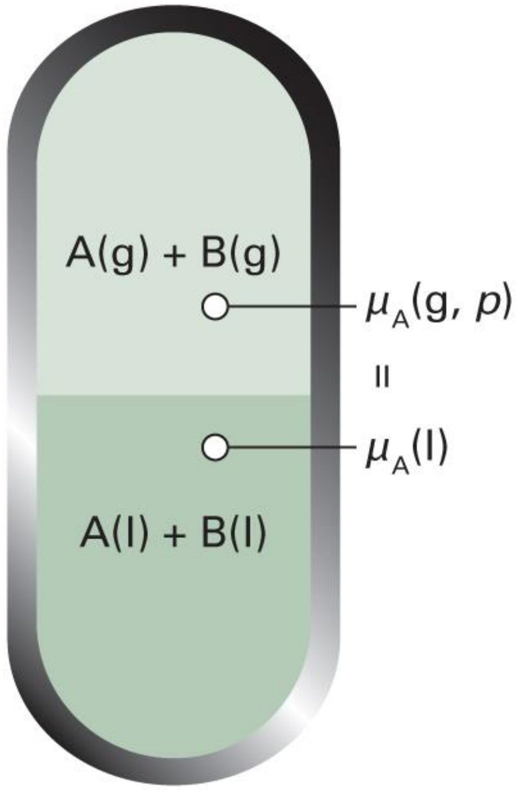
$$\underbrace{\mu_A^*(l)}_{\text{liquid}} = \underbrace{\mu_A^\ominus(g) + RT\ln(p_A^*/p^\ominus)}_{\text{(perfect) gas}}$$

If A is mixed with another substance B,

$$\mu_A(l) = \mu_A^\ominus(g) + RT\ln(p_A/p^\ominus)$$

Subtraction leads to

$$\mu_A(l) = \mu_A^*(l) + RT\ln(p_A/p_A^*)$$



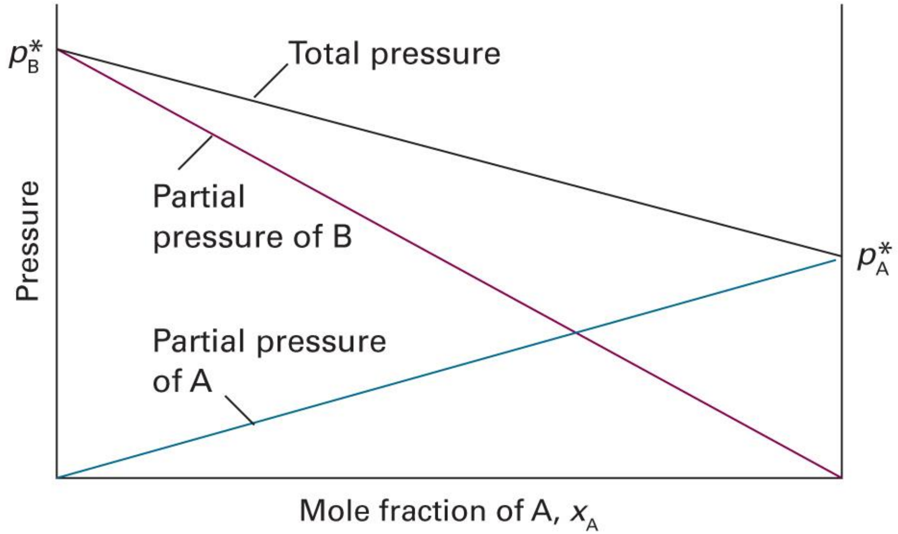
(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.10)

Review

Raoult's Law: Ideal Solution

$$p_A = x_A p_A^*$$

- Ideal solutions: mixtures that obey the law throughout the composition range from pure A to pure B



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.11)

Mixing of Ideal Solutions

$$\Delta_{\text{mix}}G = nRT(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})$$

- Entropy of mixing

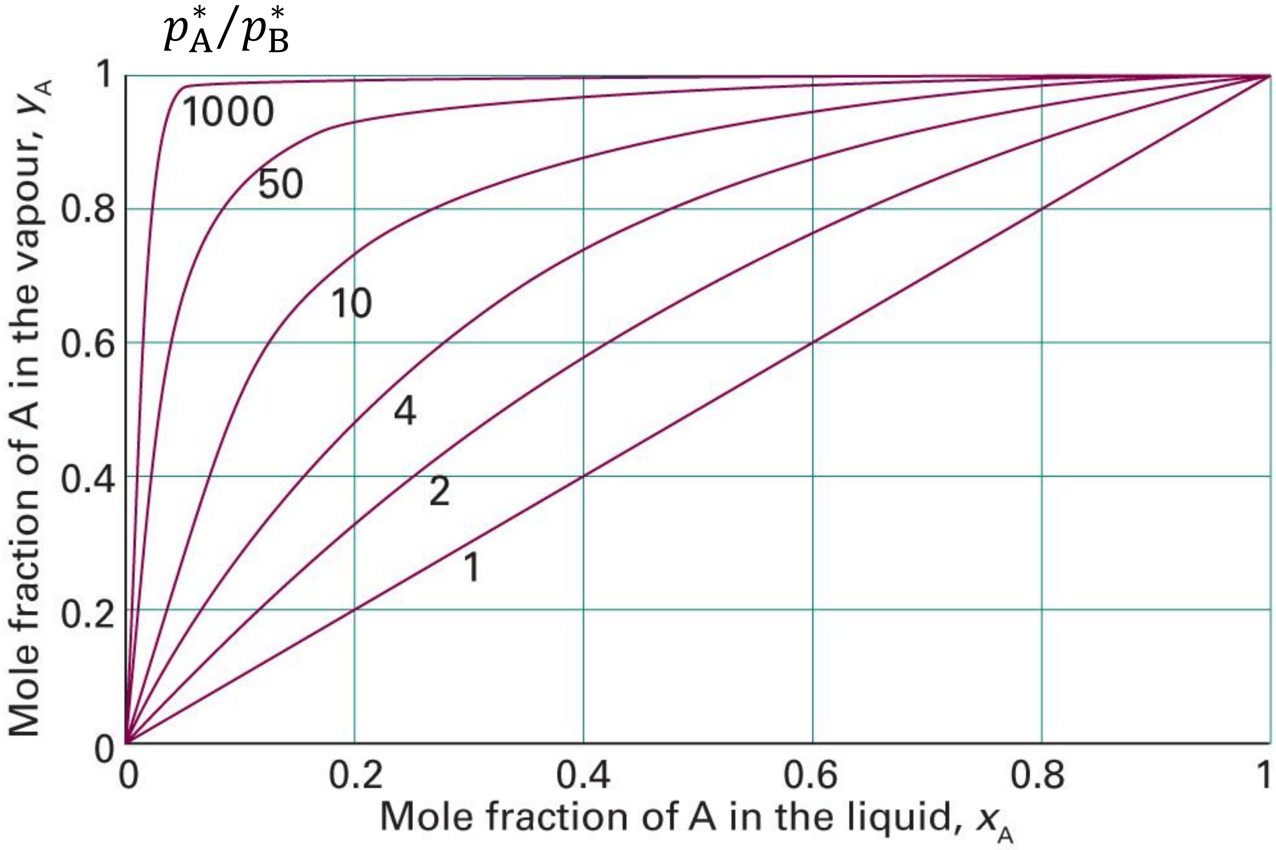
$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

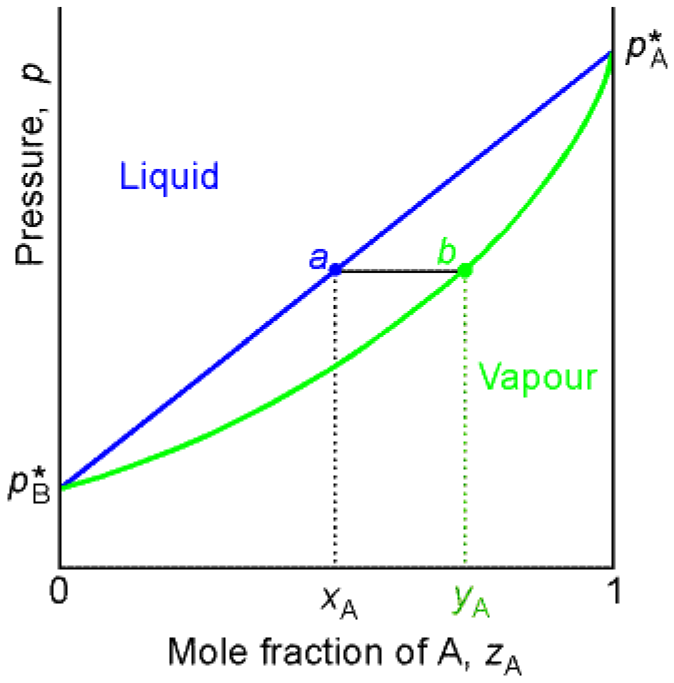
(Same with perfect gas mixtures)

Vapor Compositions



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.2)

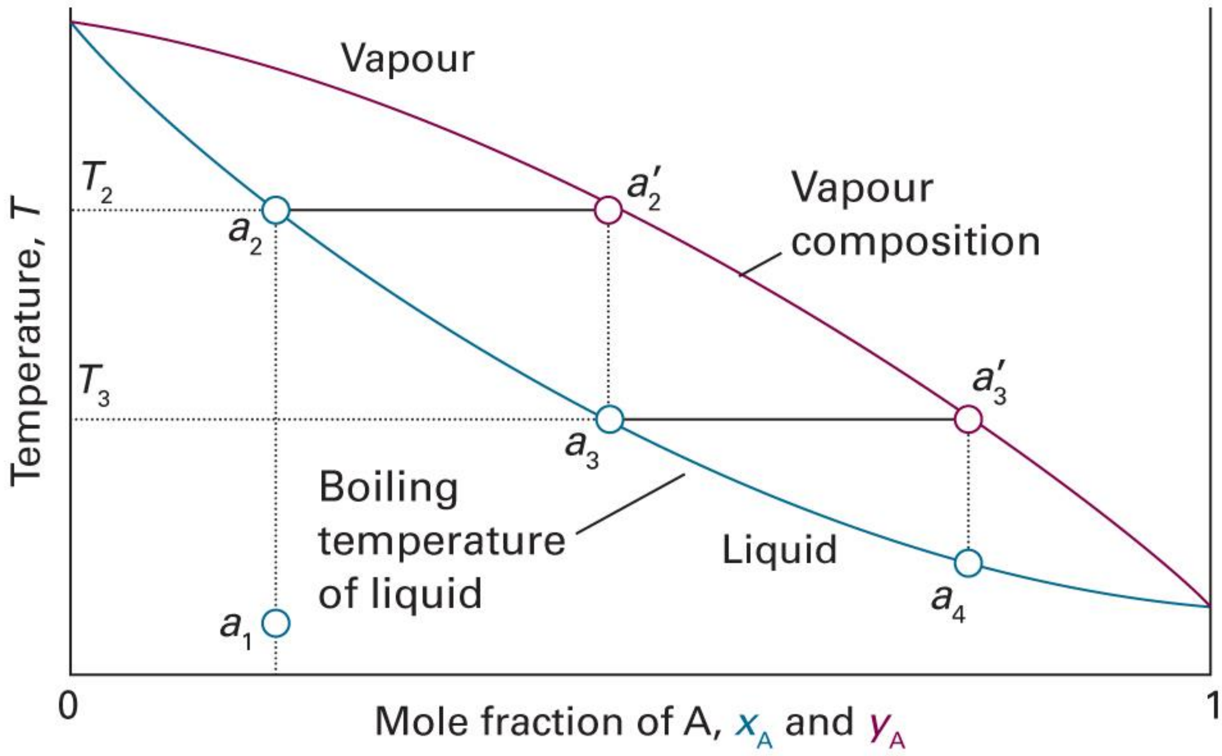
Pressure-Composition Diagram



(Image: http://people.cst.cmich.edu/teckl1mm/pchemi/chm351ch8af01.htm)

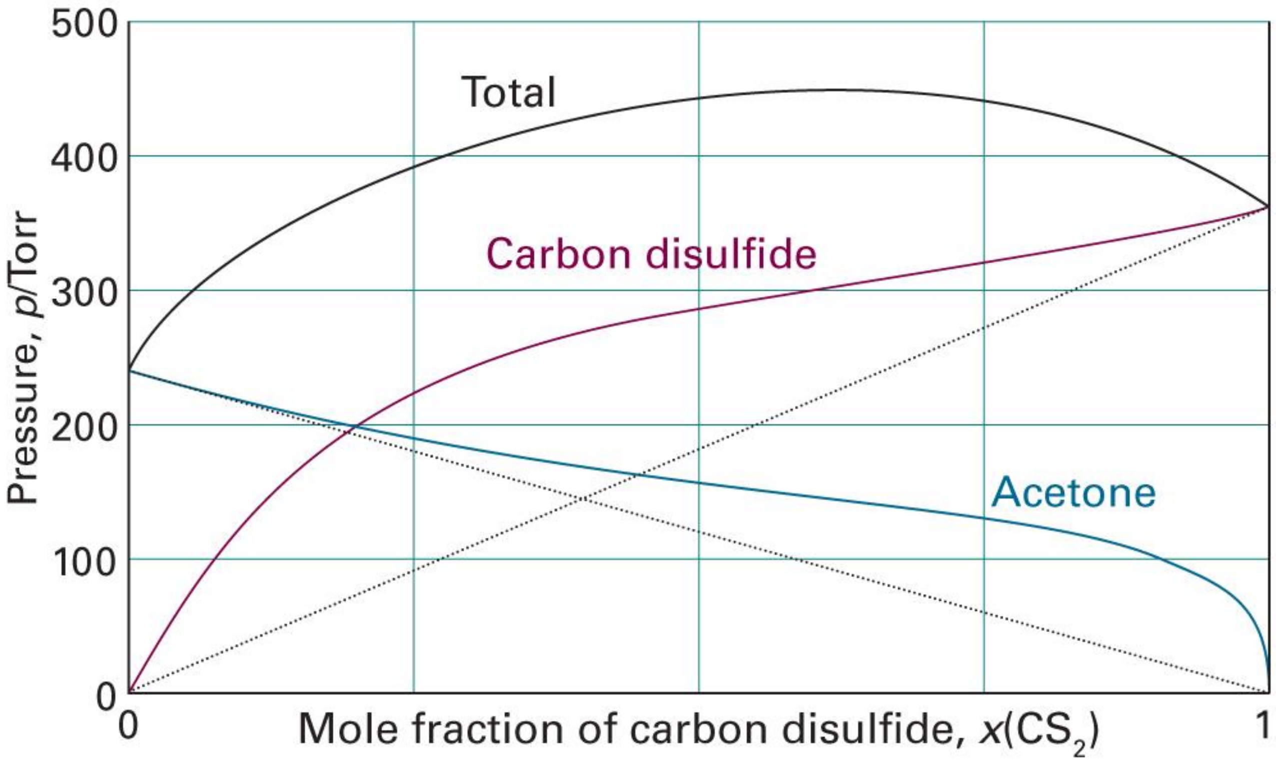
Temperature-Composition Diagram

$p_{\mathrm{A}}^{*} > p_{\mathrm{B}}^{*}$ (A is more volatile)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.4)

Real Solutions: Deviations



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.12)

Henry's Law

$$p_{\mathrm{B}} = x_{\mathrm{B}} K_{\mathrm{B}}$$

- Ideal-dilute solutions: mixtures for which the solute B obeys Henry's law and the solvent A obeys Raoult's law

The molality, b, is often instead used:

$$p_B = b_{\mathrm{B}} K_{\mathrm{B}}$$

Brief Illustration 5A.4

Estimate the molar solubility of oxygen in water at 25°C and a partial pressure of 21 kPa. (Henry's law constant of oxygen is K_{O_2} = 7.92×10^4 kPa kg mol^{-1}.)

The molality of the saturated solution is

$b_{O_2} = p_{O_2}/K_{O_2}$ = (21 kPa)/(7.92×10^4 kPa kg mol^{-1}) = 2.9×10^{-4} mol kg^{-1}

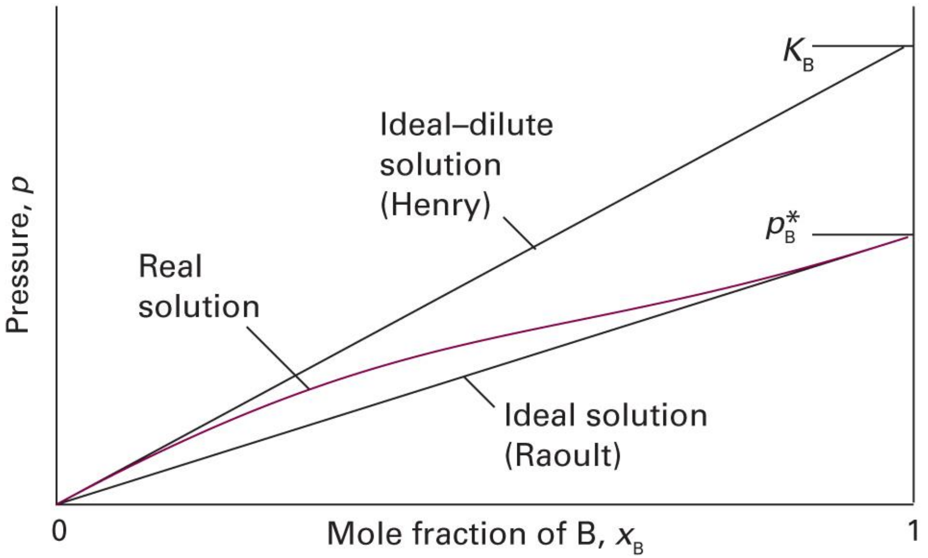
Using the mass density of pure water, ρ = 0.997 kg L^{-1},

[O_2] = $b_{O_2}\rho$ = (2.9×10^{-4} mol kg^{-1})×(0.997 kg L^{-1}) = 0.29 mM

Topic 5A

Limiting Laws for Real Solutions

- Raoult's law: reliable as $x \rightarrow 1$
- Henry's law: reliable as $x \rightarrow 0$



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.13)

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Connecting the Two Laws

$$\mu_{\mathrm{A}}(T,p) = \mu_{\mathrm{A}}^{*}(T) + RT\ln(p_{\mathrm{A}}/p_{\mathrm{A}}^{*})$$

$$\mu_{\mathrm{A}}(T,p) = \mu_{\mathrm{A}}^{*}(T) + RT\ln p_{\mathrm{A}} - RT\ln p_{\mathrm{A}}^{*}$$

Similarly,

$$\mu_{\mathrm{B}}(T,p) = \mu_{\mathrm{B}}^{*}(T) + RT\ln p_{\mathrm{B}} - RT\ln p_{\mathrm{B}}^{*}$$

Hence,

$$d\mu_{\mathrm{A}} = RT\left(\frac{\partial \ln p_{\mathrm{A}}}{\partial x_{\mathrm{A}}}\right)_{T,p} dx_{\mathrm{A}}, \qquad d\mu_{\mathrm{B}} = RT\left(\frac{\partial \ln p_{\mathrm{B}}}{\partial x_{\mathrm{B}}}\right)_{T,p} dx_{\mathrm{B}}$$

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Connecting the Two Laws

$$d\mu_{\mathrm{A}} = RT\left(\frac{\partial \ln p_{\mathrm{A}}}{\partial x_{\mathrm{A}}}\right)_{T,p} dx_{\mathrm{A}}, \qquad d\mu_{\mathrm{B}} = RT\left(\frac{\partial \ln p_{\mathrm{B}}}{\partial x_{\mathrm{B}}}\right)_{T,p} dx_{\mathrm{B}}$$

At constant T and p,

$$x_{\mathrm{A}} d\mu_{\mathrm{A}} + x_{\mathrm{B}} d\mu_{\mathrm{B}} = 0 \text{ (Gibbs-Duhem equation)}$$

Substitution leads to

$$x_{\mathrm{A}}\left(\frac{\partial \ln p_{\mathrm{A}}}{\partial x_{\mathrm{A}}}\right)_{T,p} dx_{\mathrm{A}} + x_{\mathrm{B}}\left(\frac{\partial \ln p_{\mathrm{B}}}{\partial x_{\mathrm{B}}}\right)_{T,p} dx_{\mathrm{B}} = 0$$

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Connecting the Two Laws

$$x_{\mathrm{A}}\left(\frac{\partial \ln p_{\mathrm{A}}}{\partial x_{\mathrm{A}}}\right)_{T,p} dx_{\mathrm{A}} + x_{\mathrm{B}}\left(\frac{\partial \ln p_{\mathrm{B}}}{\partial x_{\mathrm{B}}}\right)_{T,p} dx_{\mathrm{B}} = 0$$

Since $x_{\mathrm{A}} + x_{\mathrm{B}} = 1$, $dx_{\mathrm{A}} = -dx_{\mathrm{B}}$ and

$$\left\{x_{\mathrm{A}}\left(\frac{\partial \ln p_{\mathrm{A}}}{\partial x_{\mathrm{A}}}\right)_{T,p} - x_{\mathrm{B}}\left(\frac{\partial \ln p_{\mathrm{B}}}{\partial x_{\mathrm{B}}}\right)_{T,p}\right\} dx_{\mathrm{A}} = 0$$

$$x_{\mathrm{A}}\left(\frac{\partial \ln p_{\mathrm{A}}}{\partial x_{\mathrm{A}}}\right)_{T,p} = x_{\mathrm{B}}\left(\frac{\partial \ln p_{\mathrm{B}}}{\partial x_{\mathrm{B}}}\right)_{T,p}$$

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Connecting the Two Laws

$$x_\mathrm{A}\left(\frac{\partial \ln p_\mathrm{A}}{\partial x_\mathrm{A}}\right)_{T,p} = x_\mathrm{B}\left(\frac{\partial \ln p_\mathrm{B}}{\partial x_\mathrm{B}}\right)_{T,p}$$

If species A obeys Raoult's law, $p_\mathrm{A} = p_\mathrm{A}^* x_\mathrm{A}$,

$$x_\mathrm{A}\left(\frac{\partial \ln(p_\mathrm{A}^* x_\mathrm{A})}{\partial x_\mathrm{A}}\right)_{T,p} = x_\mathrm{A} \times \frac{1}{x_\mathrm{A}} = 1$$

$$x_\mathrm{B}\left(\frac{\partial \ln p_\mathrm{B}}{\partial x_\mathrm{B}}\right)_{T,p} = 1$$

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Connecting the Two Laws

$$x_B \left(\frac{\partial \ln p_B}{\partial x_B}\right)_{T,p} = 1$$

At constant T and p,

$$d \ln p_B = \frac{1}{x_B} dx_B$$

$$\int d \ln p_B = \int \frac{1}{x_B} dx_B$$

$$\ln p_B = \ln x_B + C$$

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Connecting the Two Laws

$$\ln p_{\mathrm{B}} = \ln x_{\mathrm{B}} + C$$

$$p_{\mathrm{B}} = x_{\mathrm{B}} e^{C}$$

Denoting $e^{C} = K_{\mathrm{B}}$, we obtain Henry's law:

$$p_{\mathrm{B}} = K_{\mathrm{B}} x_{\mathrm{B}}$$

Chemical Potentials

- Solvent ($x_{\mathrm{A}} \rightarrow 1$): Raoult's law

$$\mu_{\mathrm{A}}(l) = \mu_{\mathrm{A}}^{*}(l) + RT\ln(p_{\mathrm{A}}/p_{\mathrm{A}}^{*}) = \mu_{\mathrm{A}}^{*}(l) + RT\ln x_{\mathrm{A}}$$

- Solute ($x_{\mathrm{B}} \rightarrow 0$): Henry's law

$$\mu_{\mathrm{B}}(l) = \mu_{\mathrm{B}}^{*}(l) + RT\ln(p_{\mathrm{B}}/p_{\mathrm{B}}^{*}) = \mu_{\mathrm{B}}^{*}(l) + RT\ln(K_{\mathrm{B}}x_{\mathrm{B}}/p_{\mathrm{B}}^{*})$$

$$= \underbrace{\mu_{\mathrm{B}}^{*}(l) + RT\ln(K_{\mathrm{B}}/p_{\mathrm{B}}^{*})}_{\mu_{\mathrm{B}}^{\ominus}(l)} + RT\ln x_{\mathrm{B}}$$

$$= \mu_{\mathrm{B}}^{\ominus}(l) + RT\ln x_{\mathrm{B}}$$

Brief Illustration 5F.2

In a mixture of propanone and trichloromethane at 298 K, $K_{\text{propanone}}$ = 24.5 kPa, whereas $p^*_{\text{propanone}}$ = 46.3 kPa. What is the difference between $\mu^{\ominus}_{\text{propanone}}$ and $\mu^*_{\text{propanone}}$?

From the definition,

$$\mu^{\ominus}_{\text{propanone}} = \mu^*_{\text{propanone}} + RT\ln\left(K_{\text{propanone}}/p^*_{\text{propanone}}\right)$$

$$= \mu^*_{\text{propanone}} + (8.3145\ \text{J K}^{-1}\ \text{mol}^{-1}) \times (298\ \text{K}) \times \ln(24.5/46.3)$$

$$= \mu^*_{\text{propanone}} - 1.58\ \text{kJ mol}^{-1}$$

Hence, the difference is $-1.58\ \text{kJ mol}^{-1}$.

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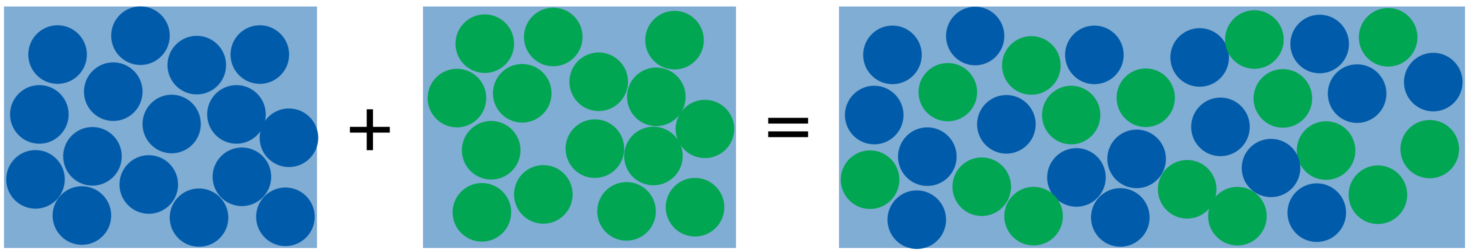
Microscopic Model

Pure substances

Mixture



	Pure substance		Mixture
Number of A molecules	N_A	0	N_A
Number of B molecules	0	N_B	N_B
Total number of molecules	N_A	N_B	$N_A + N_B$

Calculate the chemical potentials

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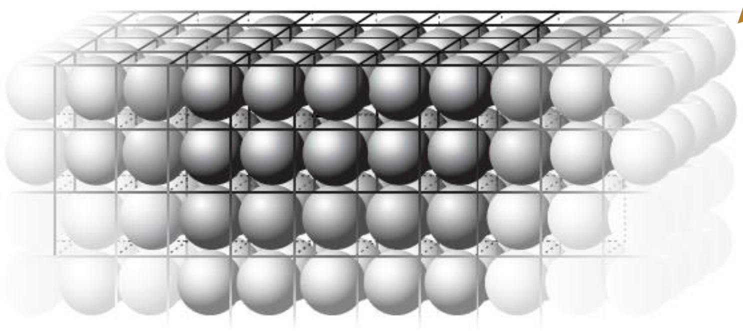
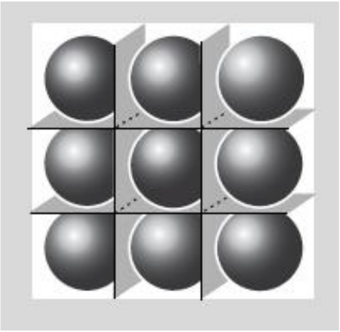
Lattice Model

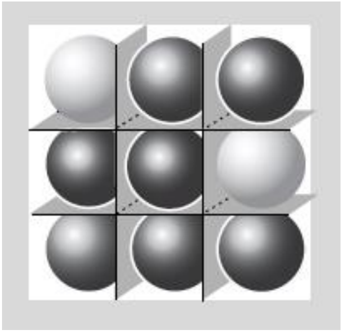
Molecules



Pure substance



Mixture



(Image: Dill and Bromberg, ***Molecular Driving Forces***, 2^{nd} Ed., Figures 14.1 and 16.2)

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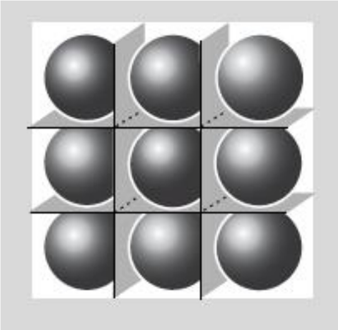
Numbers of Microstates

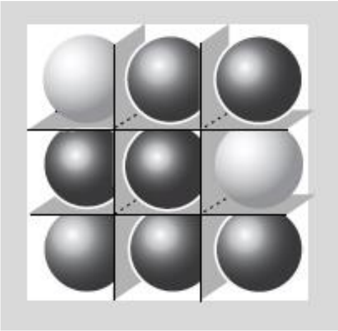
Pure



- Number of rooms: N
- Number of molecules: N

$$W = 1$$

Mixed



- Number of rooms: $N_\mathrm{A} + N_\mathrm{B}$
- Number of A molecules: N_A
- Number of B molecules: N_B

$$W = {}_{N_\mathrm{A}+N_\mathrm{B}}C_{N_\mathrm{A}} = (N_\mathrm{A} + N_\mathrm{B})!/(N_\mathrm{A}!\,N_\mathrm{B}!)$$

(Image: Dill and Bromberg, ***Molecular Driving Forces***, 2nd Ed., Figure 16.2)

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Entropy of Mixing

- Pure A: $S_{\mathrm{A}} = k_{\mathrm{B}} \ln W_{\mathrm{A}} = k_{\mathrm{B}} \ln 1 = 0$
- Pure B: $S_{\mathrm{B}} = k_{\mathrm{B}} \ln W_{\mathrm{B}} = k_{\mathrm{B}} \ln 1 = 0$
- Mixture:

$$S_{\mathrm{A+B}} = k_{\mathrm{B}} \ln W_{\mathrm{A+B}} = k_{\mathrm{B}} \ln \frac{(N_{\mathrm{A}} + N_{\mathrm{B}})!}{N_{\mathrm{A}}!\, N_{\mathrm{B}}!}$$

- Entropy of mixing

$$\Delta_{\mathrm{mix}} S = S_{\mathrm{A+B}} - S_{\mathrm{A}} - S_{\mathrm{B}} = k_{\mathrm{B}} \ln \frac{(N_{\mathrm{A}} + N_{\mathrm{B}})!}{N_{\mathrm{A}}!\, N_{\mathrm{B}}!}$$

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Stirling's Approximation

$\ln N! \approx N \ln N - N$ (for large N)

$$\begin{aligned}
\ln N! &= \ln\{N \times (N-1) \times (N-2) \times \cdots\} \\
&= \ln N + \ln(N-1) + \ln(N-2) + \cdots \\
&= \textstyle\sum_{k=1}^{N} \ln k \\
&\approx \int_1^N \ln x \, dx \\
&= [x \ln x - x]_1^N \\
&= N \ln N - N + 1 \\
&\approx N \ln N - N \text{ (for large } N \gg 1\text{)}
\end{aligned}$$

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Entropy of Mixing

$$\Delta_{\text{mix}}S = k_{\text{B}} \ln \frac{(N_{\text{A}} + N_{\text{B}})!}{N_{\text{A}}!\, N_{\text{B}}!}$$

Applying Stirling's approximation,

$$\begin{aligned}
\Delta_{\text{mix}}S &= k_{\text{B}}\{\ln(N_{\text{A}} + N_{\text{B}})! - \ln N_{\text{A}}! - \ln N_{\text{B}}!\} \\
&= k_{\text{B}}\{(N_{\text{A}} + N_{\text{B}}) \ln(N_{\text{A}} + N_{\text{B}}) - N_{\text{A}} \ln N_{\text{A}} - N_{\text{B}} \ln N_{\text{B}}\} \\
&= -k_{\text{B}} \left(N_{\text{A}} \ln \frac{N_{\text{A}}}{N_{\text{A}}+N_{\text{B}}} + N_{\text{B}} \ln \frac{N_{\text{B}}}{N_{\text{A}}+N_{\text{B}}}\right) \\
&= -Nk_{\text{B}}(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}}) \\
&= -nR(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})
\end{aligned}$$

: ideal solution entropy!

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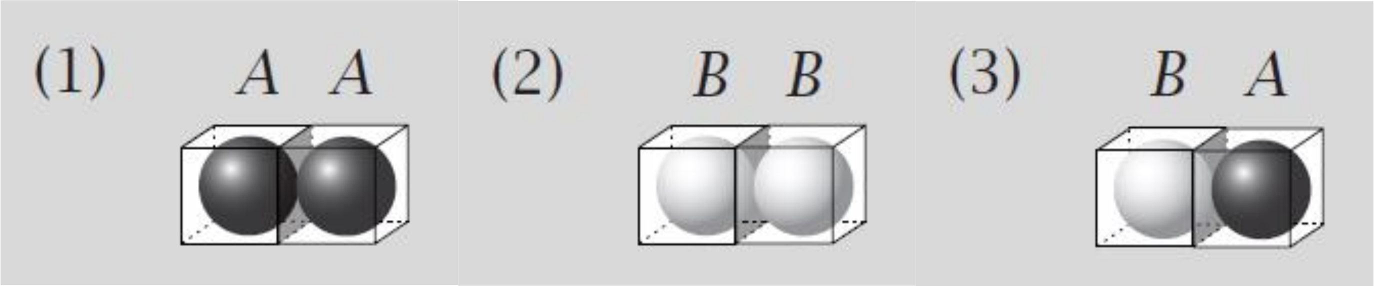
Energy of Mixture



Considering three types of contacts,

$$U = m_{\mathrm{AA}} w_{\mathrm{AA}} + m_{\mathrm{BB}} w_{\mathrm{BB}} + m_{\mathrm{AB}} w_{\mathrm{AB}}$$

where m_{ij} is the number of i-j contacts, and

w_{ij} is the energy of each i-j contact

(Image: Dill and Bromberg, ***Molecular Driving Forces***, 2nd Ed., Figure 15.4)

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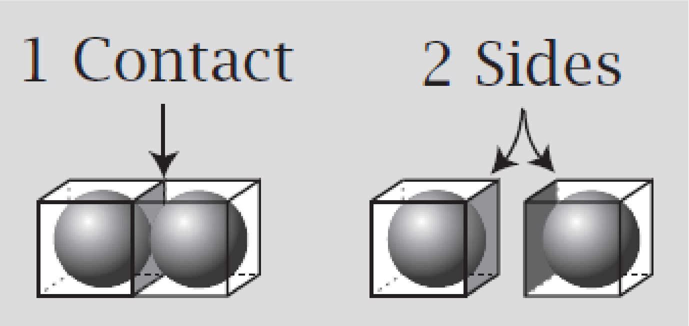
Contacts and Sides



- Each contact contributes 2 sides
- Each molecule contributes z sides
- Total number of "A" sides = $zN_{\mathrm{A}} = 2m_{\mathrm{AA}} + m_{\mathrm{AB}}$
- Total number of "B" sides = $zN_{\mathrm{B}} = 2m_{\mathrm{BB}} + m_{\mathrm{AB}}$

(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figure 15.5)

Energy of Mixture

$$zN_{\mathrm{A}} = 2m_{\mathrm{AA}} + m_{\mathrm{AB}}, \qquad zN_{\mathrm{B}} = 2m_{\mathrm{BB}} + m_{\mathrm{AB}}$$

Rearrangement leads to

$$m_{\mathrm{AA}} = (zN_{\mathrm{A}} - m_{\mathrm{AB}})/2$$

$$m_{\mathrm{BB}} = (zN_{\mathrm{B}} - m_{\mathrm{AB}})/2$$

which gives

$$U = \frac{zN_{\mathrm{A}} - m_{\mathrm{AB}}}{2} w_{\mathrm{AA}} + \frac{zN_{\mathrm{B}} - m_{\mathrm{AB}}}{2} w_{\mathrm{BB}} + m_{\mathrm{AB}} w_{\mathrm{AB}}$$

$$U = \frac{zw_{\mathrm{AA}}}{2} N_{\mathrm{A}} + \frac{zw_{\mathrm{BB}}}{2} N_{\mathrm{B}} + \left(w_{\mathrm{AB}} - \frac{w_{\mathrm{AA}} + w_{\mathrm{BB}}}{2}\right) m_{\mathrm{AB}}$$

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What is $\boldsymbol{m}_{\mathbf{AB}}$?

The probability to have B in any site:

$$p_{\mathrm{B}} = N_{\mathrm{B}}/N$$

The average number of AB contacts that each A makes:

$$zp_{\mathrm{B}} = zN_{\mathrm{B}}/N$$

The total number of AB contacts:

$$m_{\mathrm{AB}} \approx N_{\mathrm{A}} \times (zN_{\mathrm{B}}/N) = zN_{\mathrm{A}}N_{\mathrm{B}}/N$$

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Energy of Mixture

$$U = \frac{zw_{\mathrm{AA}}}{2}N_{\mathrm{A}} + \frac{zw_{\mathrm{BB}}}{2}N_{\mathrm{B}} + \left(w_{\mathrm{AB}} - \frac{w_{\mathrm{AA}} + w_{\mathrm{BB}}}{2}\right)\frac{zN_{\mathrm{A}}N_{\mathrm{B}}}{N}$$

$$U = \frac{zw_{\mathrm{AA}}}{2}N_{\mathrm{A}} + \frac{zw_{\mathrm{BB}}}{2}N_{\mathrm{B}} + \xi k_{\mathrm{B}}T\frac{N_{\mathrm{A}}N_{\mathrm{B}}}{N}$$

where the exchange parameter ξ is

$$\xi = \frac{z}{k_{\mathrm{B}}T}\left(w_{\mathrm{AB}} - \frac{w_{\mathrm{AA}} + w_{\mathrm{BB}}}{2}\right)$$

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Energy of Mixing

$$U = \frac{zw_{\mathrm{AA}}}{2} N_{\mathrm{A}} + \frac{zw_{\mathrm{BB}}}{2} N_{\mathrm{B}} + \xi k_{\mathrm{B}} T \frac{N_{\mathrm{A}} N_{\mathrm{B}}}{N}$$

- Pure A: $N_{\mathrm{B}} = 0$

$$U_{\mathrm{A}} = \frac{zw_{\mathrm{AA}}}{2} N_{\mathrm{A}}$$

- Pure B: $N_{\mathrm{A}} = 0$

$$U_{\mathrm{B}} = \frac{zw_{\mathrm{BB}}}{2} N_{\mathrm{B}}$$

$$\Delta_{\mathrm{mix}} U = U_{\mathrm{A+B}} - U_{\mathrm{A}} - U_{\mathrm{B}} = \xi k_{\mathrm{B}} T \frac{N_{\mathrm{A}} N_{\mathrm{B}}}{N}$$

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Helmholtz Energy of Mixing

$$\Delta_{\text{mix}}S = -nR(x_{\text{A}}\ln x_{\text{A}} + x_{\text{B}}\ln x_{\text{B}})$$

$$\Delta_{\text{mix}}U = \xi k_{\text{B}}T\frac{N_{\text{A}}N_{\text{B}}}{N} = \xi N k_{\text{B}}T x_{\text{A}}x_{\text{B}} = \xi nRT x_{\text{A}}x_{\text{B}}$$

$$\Delta_{\text{mix}}A = \Delta_{\text{mix}}U - T\Delta_{\text{mix}}S$$

$$\Delta_{\text{mix}}A = nRT(x_{\text{A}}\ln x_{\text{A}} + x_{\text{B}}\ln x_{\text{B}} + \xi x_{\text{A}}x_{\text{B}})$$

Regular solution: $\Delta_{\text{mix}}S = \Delta_{\text{mix}}S^{\text{ideal}}$ and $\Delta_{\text{mix}}U \neq 0$

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Energy of Mixing

For a regular solution,

$$\Delta_{\mathrm{mix}}U = nRT\xi x_{\mathrm{A}}x_{\mathrm{B}}$$

$$\xi = \frac{z}{k_{\mathrm{B}}T}\left(w_{\mathrm{AB}} - \frac{w_{\mathrm{AA}} + w_{\mathrm{BB}}}{2}\right)$$

- $\xi > 0$, or $w_{\mathrm{AB}} > (w_{\mathrm{AA}} + w_{\mathrm{BB}})/2$: mixing is endothermic
- $\xi < 0$, or $w_{\mathrm{AB}} < (w_{\mathrm{AA}} + w_{\mathrm{BB}})/2$: mixing is exothermic
- $\xi = 0$, or $w_{\mathrm{AB}} = (w_{\mathrm{AA}} + w_{\mathrm{BB}})/2$: ideal solution

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Energy to Enthalpy

For liquid-liquid mixing,

$$\Delta_{\text{mix}}H = H_{\text{A+B}} - H_{\text{A}} - H_{\text{B}}$$

$$H_{\text{A+B}} = U_{\text{A+B}} + (pV)_{\text{A+B}}$$

$$H_{\text{A}} = U_{\text{A}} + (pV)_{\text{A}}$$

$$H_{\text{B}} = U_{\text{B}} + (pV)_{\text{B}}$$

Hence,

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}U + (pV)_{\text{A+B}} - (pV)_{\text{A}} - (pV)_{\text{B}} \approx \Delta_{\text{mix}}U$$

$$\Delta_{\text{mix}}G = \Delta_{\text{mix}}H - T\Delta_{\text{mix}}S \approx \Delta_{\text{mix}}U - T\Delta_{\text{mix}}S = \Delta_{\text{mix}}A$$

Not in Atkins

Model Summary

- Ideal solution
  - Raoult's law: $p_{\mathrm{A}} = x_{\mathrm{A}} p_{\mathrm{A}}^{*}$
  - $\Delta_{\mathrm{mix}} G = -T\Delta_{\mathrm{mix}} S = nRT(x_{\mathrm{A}} \ln x_{\mathrm{A}} + x_{\mathrm{B}} \ln x_{\mathrm{B}})$
- Ideal-dilute solution
  - Solvent – Raoult's law
  - Solute – Henry's law: $p_{\mathrm{B}} = x_{\mathrm{B}} K_{\mathrm{B}}$
- Regular solution
  - $\Delta_{\mathrm{mix}} G = \Delta_{\mathrm{mix}} H - T\Delta_{\mathrm{mix}} S^{\mathrm{ideal}}$
  - Lattice model (Bragg-Williams model): $\Delta_{\mathrm{mix}} H = nRT\xi x_{\mathrm{A}} x_{\mathrm{B}}$

Topics 5B, 5E

Deviations from Ideality

- Excess function

$$X^{\mathrm{E}} = \Delta_{\mathrm{mix}}X - \Delta_{\mathrm{mix}}X^{\mathrm{ideal}}$$

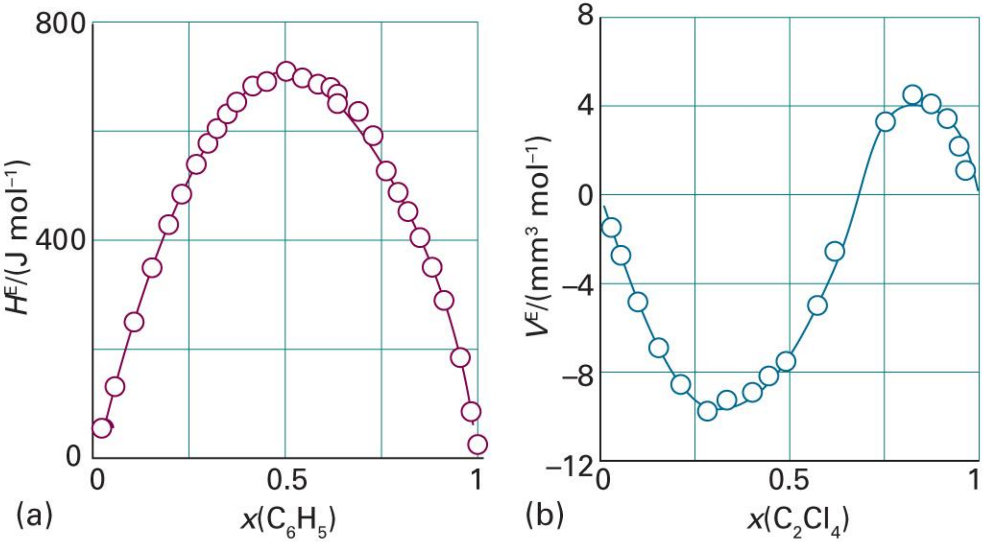
- Activity

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{*} + RT \ln a_{\mathrm{A}}$$

Excess Function

$$X^{\mathrm{E}} = \Delta_{\mathrm{mix}}X - \Delta_{\mathrm{mix}}X^{\mathrm{ideal}}$$

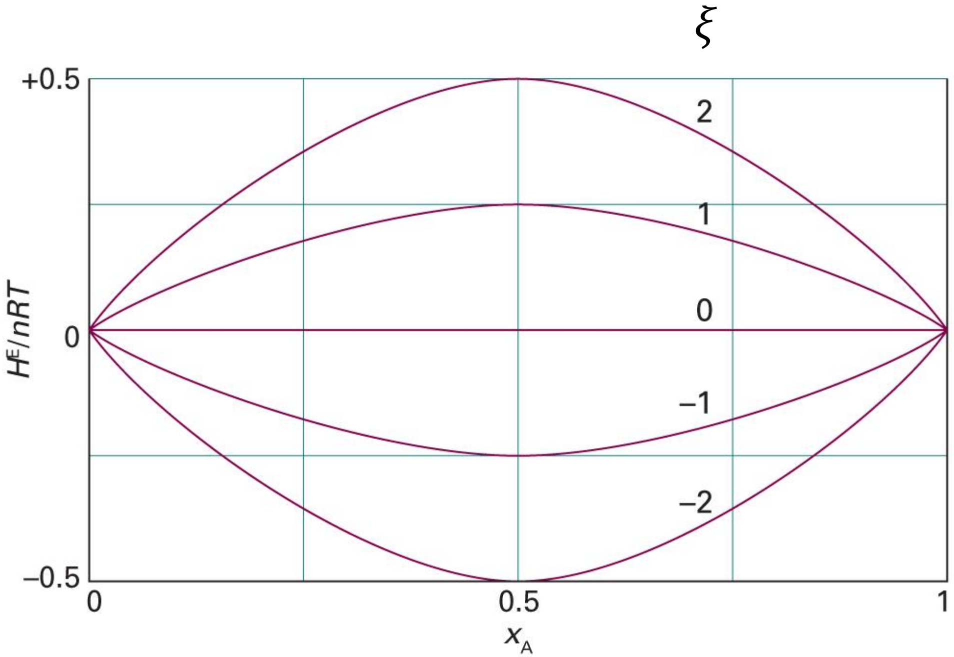
- Regular solution: $S^{\mathrm{E}} = 0$ and $U^{\mathrm{E}} \neq 0$ (or $H^{\mathrm{E}} \neq 0$)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.3)

BW Model: Excess Enthalpy

$$H^{\mathrm{E}} = \xi nRT x_{\mathrm{A}} x_{\mathrm{B}}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5B.4)

Activity a_J

- Ideal-dilute solution
  - Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln x_A$
  - Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln x_B$

    where $\mu_B^\ominus(l) = \mu_B^*(l) + RT \ln(K_B/p_B^*)$

- Real solution
  - Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln a_A$
  - Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B$

Measurement of Activity

$$\mu(l) = \mu^*(l) + RT \ln(p/p^*)$$

- Solvent

$$\mu_A(l) = \mu_A^*(l) + RT \ln a_A$$

$$a_A = p_A/p_A^*$$

- Solute

$$\mu_B(l) = \mu_B^{\ominus}(l) + RT \ln a_B = \mu_B^*(l) + RT \ln(K_B a_B/p_B^*)$$

$$a_B = p_B/K_B$$

Activity Coefficient

$$\gamma_J = a_J/x_J$$

- Solvent

$$\mu_A(l) = \mu_A^*(l) + RT\ln a_A = \mu_A^*(l) + RT\ln x_A + RT\ln\gamma_A$$

- Solute

$$\mu_B(l) = \mu_B^{\ominus}(l) + RT\ln a_B = \mu_B^{\ominus}(l) + RT\ln x_B + RT\ln\gamma_B$$

As $x_A \rightarrow 1$ and $x_B \rightarrow 0$, $a_J \rightarrow x_J$ and $\gamma_J \rightarrow 1$

Activity of BW Model

$$\begin{aligned}\Delta_{\text{mix}}G &= G - G^* = n_{\text{A}}\mu_{\text{A}} + n_{\text{B}}\mu_{\text{B}} - n_{\text{A}}\mu_{\text{A}}^* - n_{\text{B}}\mu_{\text{B}}^* \\ &= n_{\text{A}}(\mu_{\text{A}} - \mu_{\text{A}}^*) + n_{\text{B}}(\mu_{\text{B}} - \mu_{\text{B}}^*) \\ &= n_{\text{A}}RT\ln a_{\text{A}} + n_{\text{B}}RT\ln a_{\text{B}} \\ &= nRT(x_{\text{A}}\ln a_{\text{A}} + x_{\text{B}}\ln a_{\text{B}}) \\ &= nRT(x_{\text{A}}\ln x_{\text{A}} + x_{\text{B}}\ln x_{\text{B}} + x_{\text{A}}\ln\gamma_{\text{A}} + x_{\text{B}}\ln\gamma_{\text{B}})\end{aligned}$$

Comparing with the BW model Gibbs energy of mixing,

$$\Delta_{\text{mix}}G = nRT(x_{\text{A}}\ln x_{\text{A}} + x_{\text{B}}\ln x_{\text{B}} + \xi x_{\text{A}}x_{\text{B}}),$$

$$x_{\text{A}}\ln\gamma_{\text{A}} + x_{\text{B}}\ln\gamma_{\text{B}} = \xi x_{\text{A}}x_{\text{B}}$$

Activity of BW Model

$$x_A \ln \gamma_A + x_B \ln \gamma_B = \xi x_A x_B$$

$$\xi x_A x_B = \xi x_A x_B (x_A + x_B) = x_A\left(\xi x_B^2\right) + x_B\left(\xi x_A^2\right)$$

Thus, the following equations satisfy the given criterion:

$\ln \gamma_A = \xi x_B^2$, or $\gamma_A = e^{\xi x_B^2}$

$\ln \gamma_B = \xi x_A^2$, or $\gamma_B = e^{\xi x_A^2}$

(Margules equations)

BW Model and the Two Laws

$$\gamma_\mathrm{A} = e^{\xi x_\mathrm{B}^2}, \qquad \gamma_\mathrm{B} = e^{\xi x_\mathrm{A}^2}$$

For component A, the activity is

$$a_\mathrm{A} = x_\mathrm{A}\gamma_\mathrm{A} = x_\mathrm{A} e^{\xi x_\mathrm{B}^2} = x_\mathrm{A} e^{\xi(1-x_\mathrm{A})^2}$$

From the property of the activity, $a_\mathrm{A} = p_\mathrm{A}/p_\mathrm{A}^*$ and

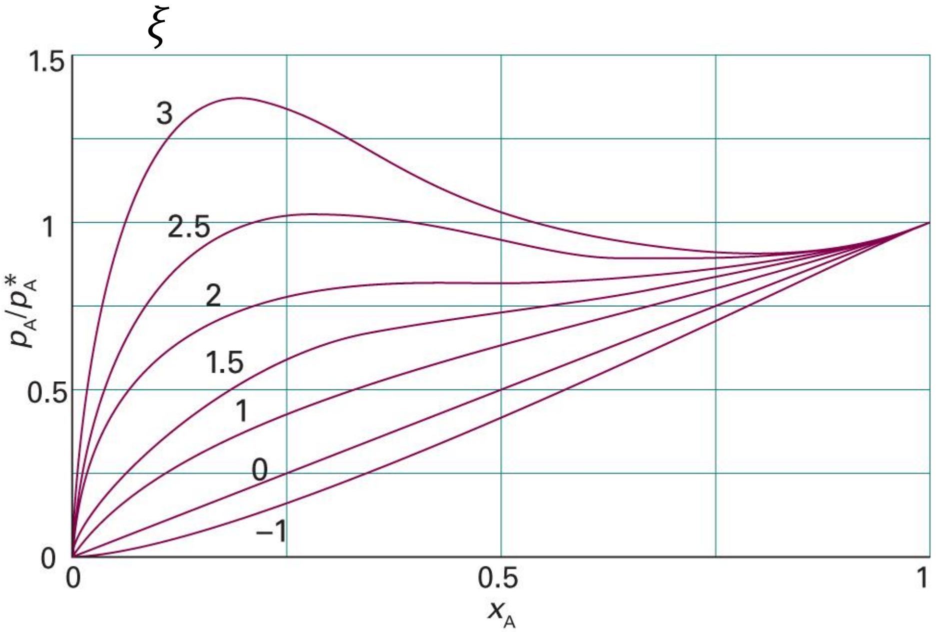
$$p_\mathrm{A} = p_\mathrm{A}^* a_\mathrm{A} = p_\mathrm{A}^* x_\mathrm{A} e^{\xi(1-x_\mathrm{A})^2}$$

- $x_\mathrm{A} \to 1$: $p_\mathrm{A} = p_\mathrm{A}^* x_\mathrm{A} e^0 = p_\mathrm{A}^* x_\mathrm{A}$ (Raoult's law)
- $x_\mathrm{A} \to 0$: $p_\mathrm{A} = p_\mathrm{A}^* x_\mathrm{A} e^\xi = K_\mathrm{A} x_\mathrm{A}$ (Henry's law)

where $K_\mathrm{A} = p_\mathrm{A}^* e^\xi$

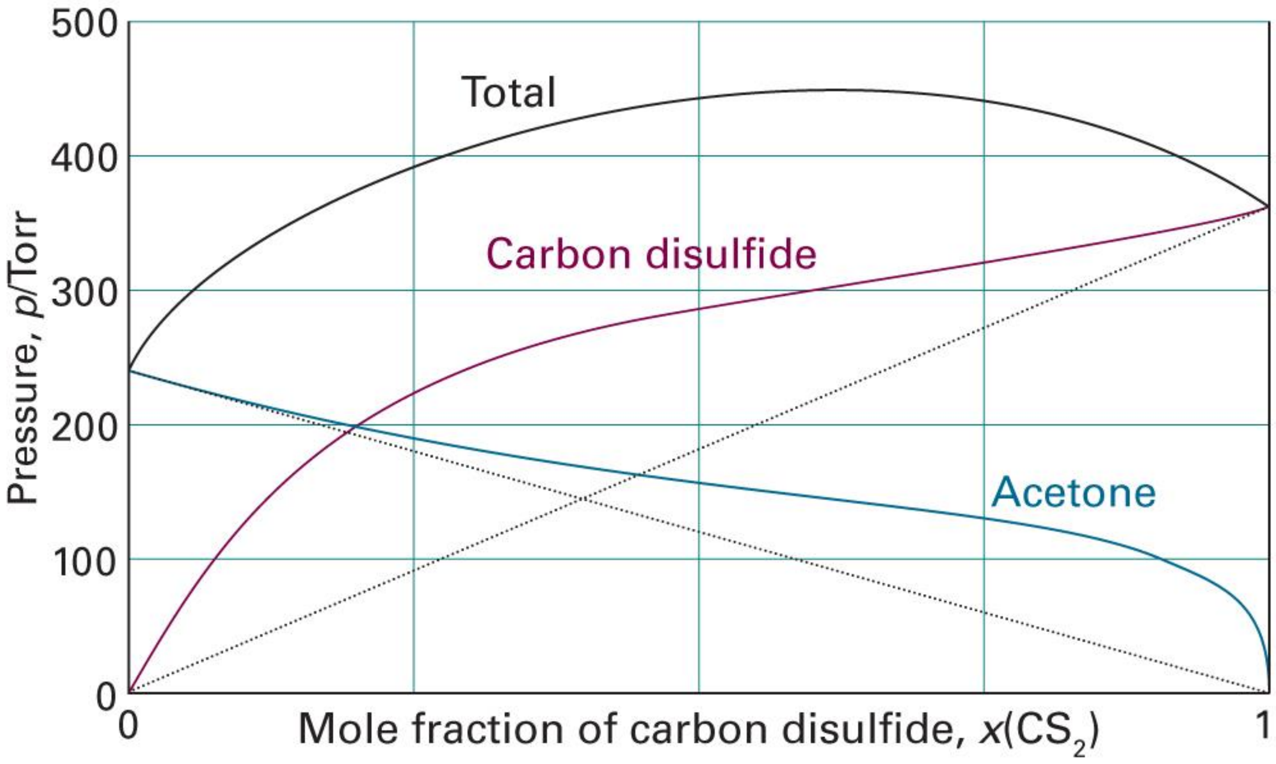
Vapor Pressure of BW Model

$$p_{\mathrm{A}}/p_{\mathrm{A}}^{*} = x_{\mathrm{A}} e^{\xi(1-x_{\mathrm{A}})^2}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5F.2)

Now We Can Explain This



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.12)